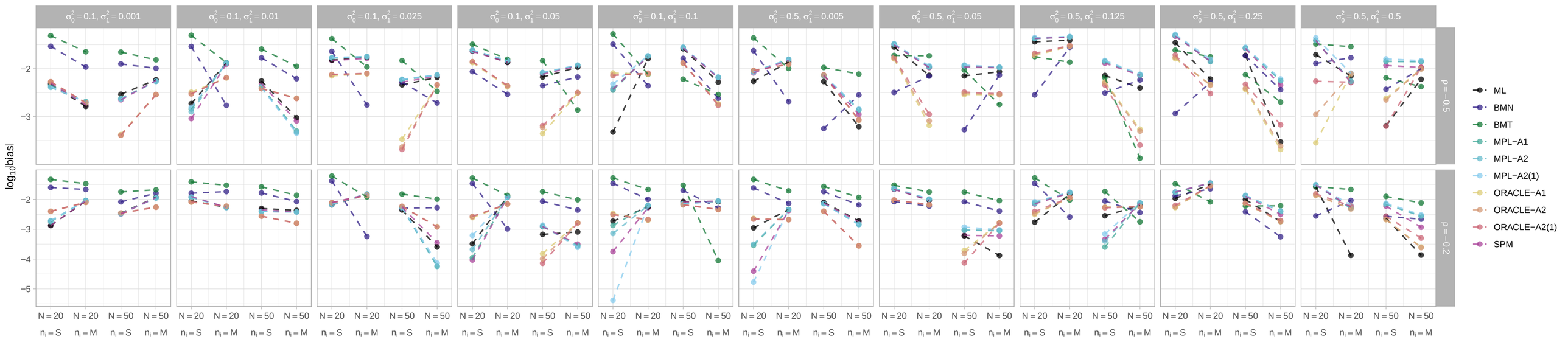
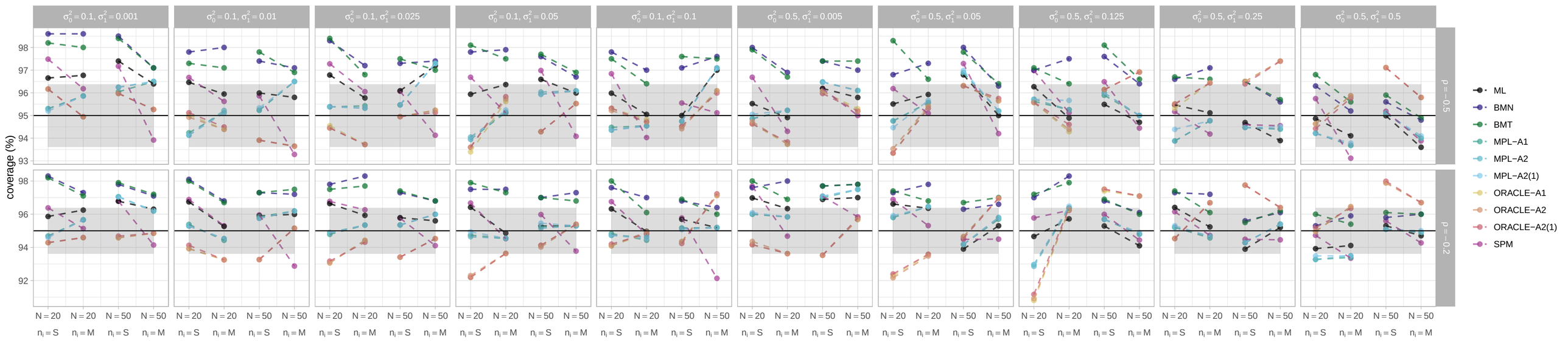

$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
$\log_{10}$|bias|
−2
−3
−2
−3
−4
−5
N = 20 N = 20 N = 50 N = 50
$n_i = S$ $n_i = M$ $n_i = S$ $n_i = M$
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM


$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
$\log_{10}$(RMSE)
−0.4
−0.5
−0.6
−0.7
−0.8
−0.9
−0.3
−0.4
−0.5
−0.6
−0.7
−0.8
−0.9
N = 20 N = 20 N = 50 N = 50
$n_i = S$ $n_i = M$ $n_i = S$ $n_i = M$
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM

$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
coverage (%)
98
97
96
95
94
93
98
96
94
92
N = 20 N = 20 N = 50 N = 50
$n_i = S$ $n_i = M$ $n_i = S$ $n_i = M$
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM

$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
log mean CI width
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
N = 20 N = 20 N = 50 N = 50
$n_i = S$ $n_i = M$ $n_i = S$ $n_i = M$
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM